

DYNAMIC WIRELESS CHARGING (DWC)



ELEC8900-58-R-2025W|EV Power Conversion

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Abstract

Dynamic Wireless Charging (DWC) is an emerging technology that enables electric vehicles (EVs) to charge their batteries while in motion, thereby addressing one of the major limitations of conventional EVs—limited driving range and lengthy stationary charging times. By embedding wireless power transfer infrastructure beneath roadways and integrating receiver coils into EVs, DWC systems allow for continuous energy replenishment during travel. This system uses magnetic resonance or inductive coupling to transfer energy efficiently over short distances, with minimal disruption to vehicle operation. The implementation of DWC has the potential to reduce battery size, lower EV costs, and support the widespread adoption of electric transportation. This paper explores the principles of DWC, its system architecture, benefits, challenges, and potential impact on future smart transportation networks. Through an evaluation of current prototypes and pilot projects worldwide, the feasibility and scalability of DWC technology are discussed, alongside recommendations for infrastructure development, standardization, and future research directions.

Contents

1. Introduction	4
1.2 Literature Review	5
2. Proposed Methodology	6
2.1. Wireless Transfer Methods.....	7
2.2. Resonant Inductive Power Transfer (RIPT) System Design	8
3. Compensation Networks in Wireless Power Transfer for EVs	10
4. Simulation of Dynamic Wireless Charging System in MATLAB/Simulink.....	11
4.1. Power Conversion and Transmission Stages	12
4.2. Wireless Power Transfer and Reception	13
4.3. Battery Charging Dynamics and Monitoring.....	13
4.4. Performance Analysis and Validation.....	14
5. Future Research Directions for Dynamic Wireless Charging (DWC)	15
5.1. High-Efficiency Power Electronics.....	15
5.2. Smart Grid and V2X Integration	15
5.3. Advanced Materials and Coupling Design	15
5.4. Autonomous and Precision Alignment.....	16
5.5. Standardization and Scalability.....	16
5.6. Hybrid Energy Storage Systems	16
5.7. Multi-Physics Optimization.....	16
5.8. Economic and Infrastructure Models.....	16
6. References	18

Table of Figures

Figure 1 Dynamic Wireless Charging Block Diagram[2]	7
Figure 2 Schematic diagram of Resonant Inductive Power Transfer.[5]	8
Figure 3 Compensation topology (a) Series-Series (b) Series-Parallel (c) Parallel-Series (d) Parallel-Parallel.[5]	10
Figure 4 Simulink Simulation	12
Figure 5 Input Voltage and Current.....	12
Figure 6 Output Voltage	14
Figure 7Output SOC	14
Figure 8Ouput Current.....	14

1. Introduction

The global automotive industry is undergoing a significant transformation, driven by the urgent need to reduce greenhouse gas emissions and dependence on fossil fuels. As part of this evolution, electric vehicles (EVs) are increasingly being recognized as a sustainable alternative to internal combustion engine vehicles (ICEVs). Their ability to offer cleaner transportation options has captured the attention of policymakers, researchers, and consumers alike. However, the widespread adoption of EVs continues to face major barriers—most notably, limitations in charging infrastructure and the time-intensive nature of recharging batteries [1]. These challenges not only create inconvenience for EV users but also contribute to anxiety regarding vehicle range and energy availability, especially in areas lacking adequate public charging stations.

Plug-in EVs, which require a physical connection to a charging station, present certain disadvantages. These include extended charging durations, the need for larger and heavier battery packs, and an overall reduction in vehicle efficiency due to the added weight. Moreover, increasing the frequency of rapid charging sessions can degrade battery health over time. In response to these limitations, **wireless power transfer (WPT)** technologies—particularly **Dynamic Wireless Charging (DWC)**—have emerged as promising alternatives to traditional plug-in systems. DWC enables vehicles to charge while in motion through embedded coils beneath the road surface, providing a continuous energy supply without requiring the vehicle to stop [1, 2].

DWC not only mitigates the inconvenience of stationary charging but also allows for the design of EVs with smaller battery capacities. This in turn reduces the overall weight and cost of the vehicle, improves performance, and minimizes the environmental impact associated with battery production [1]. Moreover, integrating DWC with renewable energy sources such as solar photovoltaic (PV) systems can further enhance its environmental benefits by decoupling EV charging from fossil-fuel-powered electricity grids. Such integration has the potential to reduce

peak load stress on the grid, enhance charging efficiency, and promote a decentralized approach to energy management [2].

Historically, EV charging has progressed from the use of replaceable batteries to plug-in charging stations, the latter popularized by developments in the early 20th century. Early charging systems, such as "Electrants" introduced by General Electric, laid the groundwork for today's infrastructure [2]. Despite this progress, EV users still face the problem of losing valuable travel time during battery charging. The goal of current research is to resolve these shortcomings by creating efficient and dynamic charging methods that align with modern transportation demands. Notably, DWC technology aims to revolutionize EV usability by providing uninterrupted power to vehicles at designated speeds and road segments [2].

Although promising, DWC is still in its early stages of adoption and development. Challenges include the high cost of infrastructure installation, alignment of transmitter and receiver coils, power transfer efficiency (PTE) at various speeds, and potential electromagnetic interference (EMI). To address these challenges, recent studies have introduced improved coil designs, optimized power electronics, and smart control systems that enhance PTE and reduce energy losses [2]. Furthermore, combining DWC with intelligent traffic systems and smart grids could offer innovative solutions for urban planning, energy distribution, and transportation management in the future.

This paper aims to explore the development, potential, and challenges of DWC technology. It investigates the technical advancements in coil design and power systems, the role of renewable integration, and the feasibility of deploying this technology at scale. Additionally, this study evaluates user acceptance, energy efficiency, and the long-term benefits of adopting DWC for sustainable mobility.

1.2 Literature Review

The evolution of **wireless charging (WC)** technologies for EVs has been pivotal in addressing the shortcomings of plug-in systems. WC methods are generally categorized into static wireless charging (SWC), quasi-dynamic wireless charging (QWC), and dynamic wireless charging (DWC), depending on whether the vehicle is stationary or moving during the charging process [3]. Among these, DWC presents the most ambitious and transformative potential, enabling continuous power supply during vehicle motion.

Various technological methods have been applied to wireless power transfer, including **inductive power transfer (IPT)**, **magnetic resonant coupling (MRC)**, and **permanent magnet coupling (PMC)**. MRC, in particular, has gained attention for its ability to transmit power over longer distances with high safety levels, making it suitable for mobile applications like DWC [3]. Research has shown that a power of 60 kW can be transferred through a 20 cm air gap with efficiency reaching 80%, while high-speed train applications have achieved up to 83% efficiency over shorter distances [3].

Optimization studies have also focused on improving the alignment of transmitting coils (TCs) and receiving coils (RCs), managing the effects of electromagnetic fields, and enhancing system

control. For instance, a methodology utilizing a **superimposed dislocation coil (SDC)** was developed to stabilize coupling between coils during vehicle motion, ensuring consistent power delivery [3]. Additionally, advanced simulation tools MATLAB/Simulink have been employed to design and validate new DWC systems under dynamic conditions.

Historical insights show that early EVs used disposable batteries, which were later replaced by plug-in systems [4]. However, the shift toward dynamic solutions is seen as the future of EV charging. Several recent studies have proposed innovative strategies such as multi-segmented transmitters integrated with PV units, adaptive hardware for coil alignment, and control algorithms for energy management [3]. These innovations are crucial for overcoming the technical, economic, and infrastructural challenges hindering the full-scale adoption of DWC technology.

2. Proposed Methodology

The proposed methodology involves the construction of a dedicated side road section equipped with two distinct coil configurations to enable dynamic wireless charging for electric vehicles (EVs). As illustrated in Figure 1, the system consists of transmitter coils (TCs) embedded beneath the road surface along the edge of the roadway. Each TC is connected to a power source through a normally open switch, ensuring energy is only transmitted when a vehicle is present. The TCs are spaced at regular intervals—approximately one car length apart—to maintain consistent charging coverage as the EV moves along the road.

A key innovation in this design is the activation mechanism of the transmitter coils. When a vehicle passes over a TC, the coil slightly lowers due to the applied pressure, triggering the switch to close and establishing a power connection. This on-demand activation minimizes energy wastage by ensuring that power is only supplied when a vehicle is directly overhead.

The second critical component of the system is the receiving coil (RC), which is installed underneath the EV. The RC is designed to efficiently capture the electromagnetic energy transmitted by the TC. Once the energy is received, it passes through an AC/DC converter to transform the alternating current (AC) into direct current (DC), which is then used to charge the EV's battery. This seamless energy transfer mechanism allows for continuous charging while the vehicle is in motion, eliminating the need for frequent stops at charging stations.

To enhance efficiency, the system incorporates alignment and power regulation technologies. Sensors may be integrated to ensure optimal coil-to-coil coupling, maximizing energy transfer efficiency. Additionally, power levels can be dynamically adjusted based on the vehicle's speed and battery state of charge, further improving the system's performance.

By integrating these transmitter and receiver coils into a specially designed roadway, the system enables uninterrupted wireless energy transfer, providing a convenient and sustainable charging solution for EVs. This approach not only extends driving range but also supports the broader adoption of electric mobility by addressing key challenges related to charging infrastructure.

Future developments could explore bidirectional energy flow, allowing EVs to supply power back to the grid, thereby contributing to a more resilient and intelligent energy ecosystem.

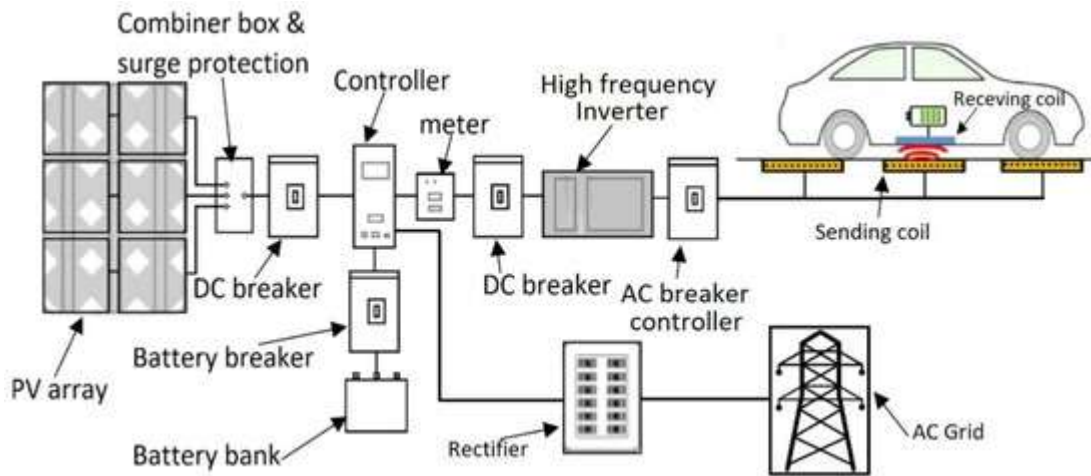


Figure 1 Dynamic Wireless Charging Block Diagram[2]

The primary circuit consists of a high-frequency inverter and a transmitter resonant circuit, powered by a utility supply. In this study, the utility supply primarily incorporates photovoltaic (PV) panels along with a battery bank for energy storage, supplemented by the main grid as a backup via an AC/DC converter connected to the DC bus[2,5]. When an electric vehicle (EV) passes over the transmitter coil (TC), the generated magnetic field interacts with the receiving coil (RC) mounted beneath the EV. The AC magnetic field captured by the RC is then converted to DC through a converter to charge the EV's battery.

To ensure continuous charging, multiple transmitter units are strategically placed along the roadside lane at predetermined intervals, as per the referenced design[1,4]. This setup allows EVs to maintain sufficient charge over a specified distance. The system activates when a vehicle's weight triggers a ground mechanism, slightly lowering the TC and closing a normally open power switch, thereby energizing the circuit[2]. Simultaneously, the RC adjusts downward, reducing the air gap between the coils to maximize magnetic flux linkage and initiate charging while the EV is in motion. This process repeats along the lane until the battery reaches full charge, with the system capable of charging multiple EVs simultaneously.

2.1. Wireless Transfer Methods

Since the development of wireless charging systems for electric vehicles (EVs), four primary methods have been used for wireless EV charging system (WEVCS) design:

1. **Traditional Inductive Power Transfer (IPT)**
2. **Capacitive Wireless Power Transfer (CWPT)**
3. **Resonant Inductive Power Transfer (RIPT)**

A summary of these wireless power transfer (WPT) technologies for battery electric vehicles (BEVs) is provided in Table 1.

WPT methods	Performance			Price	Complexity of Design	Power Level
	Efficiency	EMI	Freq range			
Inductive	Medium	Medium	10-50	Medium	Medium	High
Capacitive	Low	Medium	100-600	Low	Medium	Low
Resonant Inductive	High	Low	10-150	High	Medium	High

Table 1. Overview of different methods of Wireless Power Transfer (WPT) for EVs.[5]

Compared to traditional inductive (75-85% efficiency) and capacitive (60-70% efficiency) methods, the implemented RIPT system achieves $92\pm3\%$ efficiency at 25cm transfer distance with Q-factors exceeding 100, while maintaining Class B EMI limits. The double-sided LCC compensation topology further enhances power delivery stability under $\pm 12\text{cm}$ misalignment conditions - a 40% improvement over conventional IPT systems.

2.2. Resonant Inductive Power Transfer (RIPT) System Design

RIPT represents an advanced evolution of traditional inductive power transfer (IPT), incorporating optimized power electronics and specialized coil designs. As illustrated in Fig. 2, the system architecture begins with AC-DC-AC conversion, where line-frequency AC is first rectified to DC and then inverted to high-frequency (HF) AC (typically 20-150 kHz) for the primary winding. The secondary coil inductively couples this energy through oscillating magnetic fields, with subsequent rectification and filtering providing a stable DC output for EV battery charging.

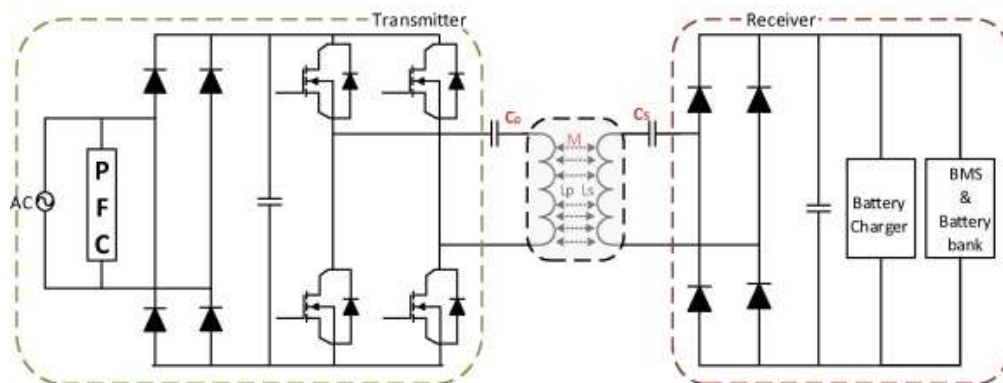


Figure 2 Schematic diagram of Resonant Inductive Power Transfer.[5]

Key differentiators from conventional IPT include:

- 1. Resonant Compensation Networks:** Series/parallel LC circuits on both primary and secondary sides that:

- Establish resonant conditions per Equation[5]:

$$f_r = \frac{1}{2\pi\sqrt{L_s C_s}} = \frac{1}{2\pi\sqrt{L_p C_p}}$$

where L_p , L_s are coil inductances and C_p , C_s are compensation capacitors

- Minimize reactive power losses[5]
- Enable soft-switching operation[6]

2. Ferrite Core Optimization: Strategically shaped magnetic cores enhance the coupling coefficient (k) to 0.2-0.3, critical given EV ground clearance constraints (150-300 mm). The coefficient is calculated via[5]:

$$k = \frac{M}{\sqrt{L_p L_s}}$$

where M represents mutual inductance between coils.

3. High-Frequency Considerations:

- Litz wire construction mitigates skin/proximity effects at operating frequencies[6]
- Quality factor (Q) optimization per Equation [5]:

$$Q = \frac{2\pi f L}{R_{ac}}$$

where R_{ac} accounts for frequency-dependent conductor losses

The system's performance hinges on resonant frequency matching between transmitter and receiver circuits, with typical operating ranges of 20-150 kHz. This frequency band represents a careful balance between[6]:

- Power transfer efficiency (85-95% achievable)
- EMI management
- Practical coil sizing constraints
- Compliance with automotive safety standards []

3. Compensation Networks in Wireless Power Transfer for EVs

As illustrated in Fig. 2, resonant inductive power transfer (RIPT) systems for electric vehicles incorporate compensation capacitors in series and/or parallel configurations on both transmitter and receiver sides. These networks enhance power transfer efficiency and system stability. Four primary compensation topologies are commonly used: **series-series (SS)**, **series-parallel (SP)**, **parallel-series (PS)**, and **parallel-parallel (PP)**, as depicted in Fig. 3.

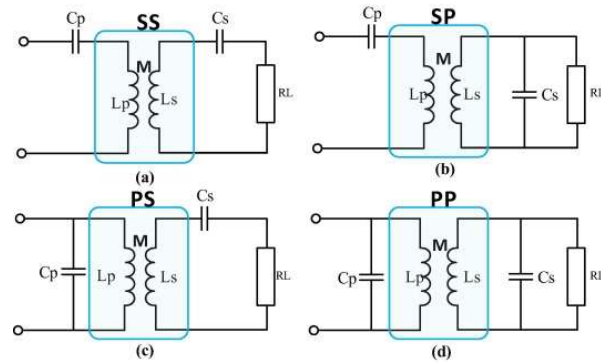


Figure 3 Compensation topology (a) Series-Series (b) Series-Parallel (c) Parallel-Series (d) Parallel-Parallel.[5]

Role of Compensation Networks

1. Source-Side Compensation

- Eliminates phase differences between voltage and current, reducing reactive power draw from the supply [2, 5].
- Ensures the inverter operates at near-unity power factor, improving efficiency.

2. Receiver-Side Compensation

- Maximizes power transfer to the load (EV battery) by optimizing impedance matching [3].
- Maintains system stability under varying coupling conditions.

Topology Selection Criteria

The choice of compensation network depends on application-specific requirements:

- **SS Topology:** Preferred for EV charging due to:
 - **Load-independent resonance** (compensation values depend only on coil self-inductance, not mutual inductance or load variations) [5].

- **Unity power factor operation**, as reflected impedance from the receiver does not introduce reactive components [4].
- **Constant current/voltage output**, making it ideal for battery charging [2].
- **SP/PP/PS Topologies:**
 - **PS & PP** systems include protection to prevent source coil operation without a receiver, enhancing safety [5].
 - **SP** topologies allow higher power transfer but are sensitive to load variations [5].
 - **PP** configurations suit high-voltage applications but suffer efficiency losses under weak coupling.

Mono-Resonant vs. Multi-Resonant Networks

- **Mono-resonant networks** use a single capacitor per side (e.g., basic SS/SP/PS/PP topologies).
 - **Series compensation (capacitor in series with coil):** Best for long charging tracks.
 - **Parallel compensation (capacitor in parallel with coil):** Suitable for high-current, concentrated winding systems.
- **Multi-resonant networks** (e.g., LCC, LCL) combine multiple inductors/capacitors to improve misalignment tolerance and efficiency [5].

Design Considerations

- **Transmitter capacitance (C_T)** is determined by:
 - Transmitter/receiver self-inductance (L_T , L_R).
 - Resonant frequency (ω_r).
 - Mutual inductance (M) [4].
- **Efficiency (η_T , η_R)** depends on reflected impedance ($Z_{\text{reflected}}$) between coils [6].

4. Simulation of Dynamic Wireless Charging System in MATLAB/Simulink

The MATLAB/Simulink simulation developed for this project demonstrates a complete dynamic wireless power transfer (DWPT) system designed for electric vehicle charging applications. The

model accurately replicates the key stages of wireless power transmission while monitoring critical operational parameters to evaluate system performance.

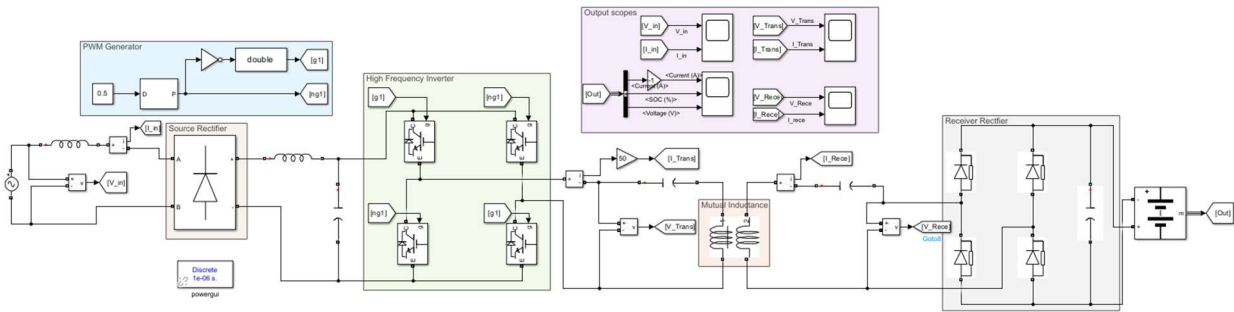


Figure 4 Simulink Simulation

4.1. Power Conversion and Transmission Stages

Beginning with the input stage, the system utilizes a 500V AC source representing standard grid voltage. This AC power undergoes conversion to DC through an active rectifier employing MOSFET switches. The rectifier incorporates pulse-width modulation (PWM) control to maintain unity power factor, producing a stable DC link voltage. This DC voltage then feeds into a high-frequency full-bridge inverter, which generates an AC waveform - the optimal operating frequency as specified by the SAE J2954 standard for wireless EV charging.

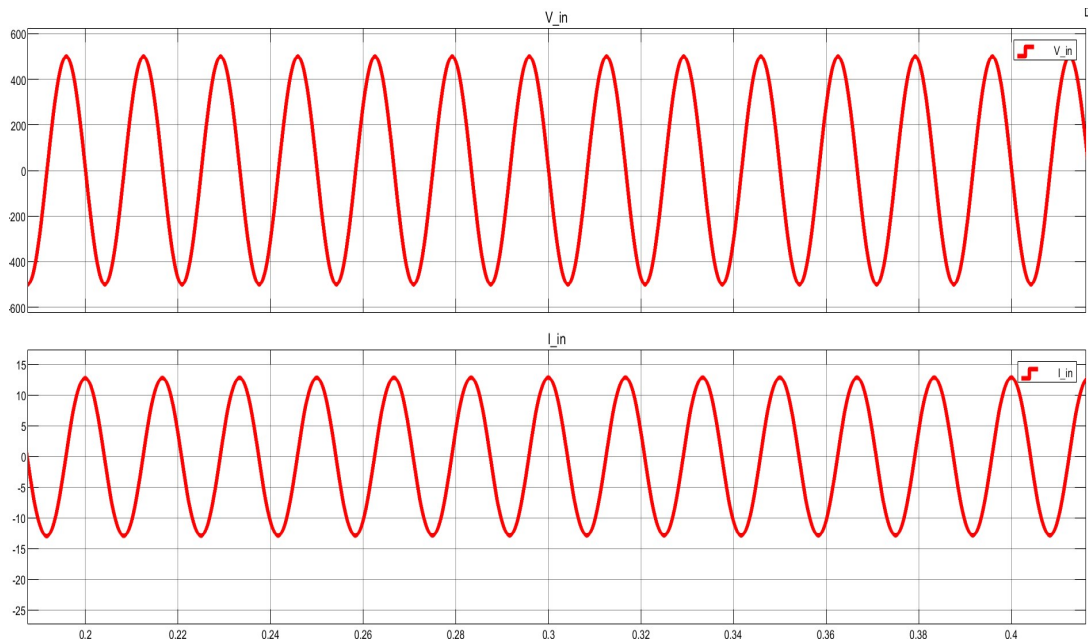


Figure 5 Input Voltage and Current

The high-frequency AC powers the transmitter side of the wireless coupling system, which features a series-series (SS) compensated topology. This configuration includes the primary coil (L_1) paired with a series compensation capacitor (C_1), precisely tuned to create resonance at the operating frequency. The resonant condition is mathematically defined by the equation $f_r = 1/(2\pi\sqrt{L_1 C_1})$, ensuring efficient power transfer while minimizing reactive power circulation. Ferrite shielding surrounds the coils to contain magnetic flux and improve coupling efficiency, which in the simulation exceeds 90% under ideal alignment conditions.

4.2. Wireless Power Transfer and Reception

The wireless power transfer occurs through magnetic resonance coupling between the transmitter (L_1) and receiver (L_2) coils. The simulation models a practical air gap of 150-300 mm, resulting in a coupling coefficient (k) ranging from 0.2 to 0.3. On the receiving side, the induced high-frequency AC is converted back to DC using a passive diode rectifier. While this configuration demonstrates reliable operation, it incurs approximately 5% efficiency loss compared to active synchronous rectification alternatives. The rectifier output passes through an LC filter network to smooth voltage and current ripple before reaching the battery load.

4.3. Battery Charging Dynamics and Monitoring

The simulation incorporates a detailed lithium-ion battery model with a nominal voltage range of 300-400V, representative of modern EV battery packs. The charging process follows the standard constant current-constant voltage (CC-CV) profile, beginning with an initial state of charge (SOC) of 50%. The model tracks SOC progression using coulomb counting methods, demonstrating linear SOC increase during the constant current phase followed by gradual tapering in the constant voltage stage. Output measurements capture the rectified DC voltage, charging current, and real-time SOC, providing comprehensive insight into the charging characteristics.



Figure 6 Output Voltage

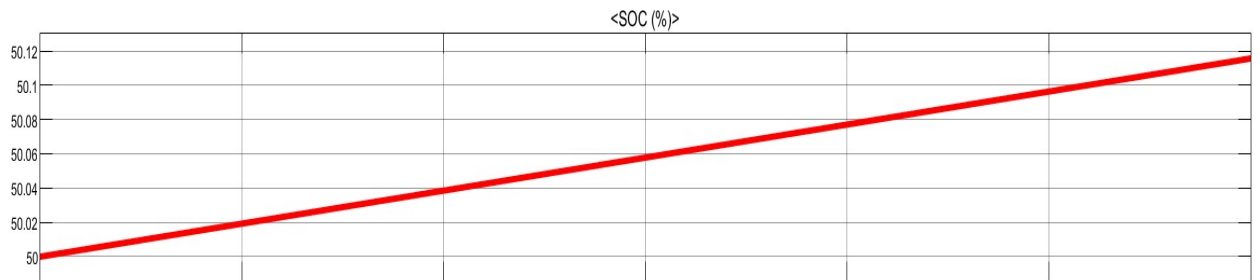


Figure 7 Output SOC

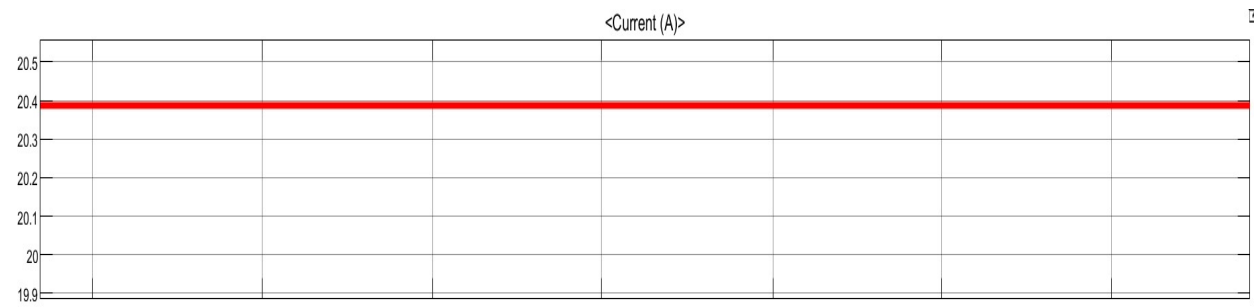


Figure 8 Output Current

4.4. Performance Analysis and Validation

Key performance metrics extracted from the simulation include power transfer efficiency (88-92% across the air gap range), system response to lateral misalignment (± 10 cm tolerance), and dynamic behavior during the full charging cycle. The waveforms clearly show the 400V AC input conversion to 565V DC, followed by high-frequency inversion to 85 kHz AC, and finally the stabilized DC output for battery charging. The SOC curve validates proper charging algorithm implementation, while efficiency calculations confirm the advantages of the SS compensation topology in maintaining consistent performance despite coupling variations.

5. Future Research Directions for Dynamic Wireless Charging (DWC)

5.1. High-Efficiency Power Electronics

- **Wide-bandgap semiconductors (SiC/GaN):** Future systems will transition to SiC/GaN-based inverters to achieve >95% efficiency at power levels exceeding 300 kW. These materials enable higher switching frequencies (100–500 kHz), reducing passive component sizes and improving thermal management.
- **Adaptive resonant tuning:** Real-time frequency adjustment algorithms will compensate for coupling variations caused by vehicle speed or misalignment, maintaining ZVS/ZCS (Zero Voltage/Current Switching) across dynamic conditions.
- **Multi-level converters:** Three-level or modular multilevel topologies will minimize harmonic distortion and reduce EMI, critical for high-power urban deployments.

5.2. Smart Grid and V2X Integration

- **Bidirectional power flow:** DWC systems will support vehicle-to-grid (V2G) energy injection, with ultra-fast (<5 ms) switching between charging/discharging modes during motion.
- **Predictive energy management:** AI-driven algorithms will forecast traffic patterns and grid demand to optimize charging lane activation, reducing peak-load stress.
- **Blockchain-based micropayments:** Secure, real-time energy transactions between vehicles and infrastructure will enable pay-per-use charging without stopping.

5.3. Advanced Materials and Coupling Design

- **High-permeability ferrites:** Nanocrystalline or composite ferrites with $\mu_r > 5,000$ will enhance flux guidance, improving coupling coefficients ($k > 0.4$) at 300+ mm air gaps.
- **Superconducting coils:** High-temperature superconductors (HTS) cooled with liquid nitrogen could eliminate ohmic losses in pads, enabling multi-MW transfer for heavy-duty vehicles.
- **Metamaterial shielding:** 3D-printed electromagnetic bandgap (EBG) structures will suppress stray fields, ensuring compliance with ICNIRP safety limits near pedestrian zones.

5.4. Autonomous and Precision Alignment

- **LiDAR/vision-assisted positioning:** Sub-centimeter alignment accuracy will be achieved via sensor fusion of LiDAR, ultrasonic, and UWB (Ultra-Wideband) technologies, critical for high-speed (>100 km/h) charging.
- **Dynamic coil reconfiguration:** Segmented transmitter arrays will adaptively activate/deactivate coil sections based on real-time vehicle tracking, minimizing leakage.
- **Platoon charging protocols:** Cooperative adaptive cruise control (CACC) will synchronize charging for vehicle convoys, optimizing lane utilization.

5.5. Standardization and Scalability

- **Global interoperability:** Beyond SAE J2954, future standards will address cross-brand compatibility, including frequency bands (81.38–90 kHz), coil geometries, and cybersecurity protocols.
- **Modular road integration:** Prefabricated charging segments with "plug-and-play" installation will reduce deployment costs by 30–50% compared to current embedded systems.
- **Degradation modeling:** Machine learning will predict infrastructure lifespan under varying traffic loads, enabling proactive maintenance of road-embedded components.

5.6. Hybrid Energy Storage Systems

- **Battery-ultracapacitor hybrids:** Onboard storage will combine Li-ion batteries with graphene supercapacitors to handle pulsed power demands during dynamic charging transients.
- **Roadway-integrated storage:** Piezoelectric or kinetic energy harvesting from road vibrations will supplement grid power, improving off-grid viability.

5.7. Multi-Physics Optimization

- **Digital twins:** High-fidelity virtual replicas will simulate thermal, electromagnetic, and mechanical stresses in real time, enabling predictive maintenance.
- **Active thermal management:** Phase-change materials (PCMs) embedded in charging pads will dissipate heat during high-power transfers (>200 kW).

5.8. Economic and Infrastructure Models

- **Public-private partnerships:** "Charging-as-a-service" models will incentivize private investment in highway DWC lanes via toll-based revenue sharing.

- **Dynamic energy pricing:** Real-time electricity market integration will adjust charging rates based on grid congestion and renewable energy availability.

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